Binomial-mixture likelihood

Parametrisation

Details

This likelihood is a mixture of three binomials, where two of them are simple predictors and one is general. The probability for success is

$$p = w_1 p_1 + w_2 p_2 + w_3 p_3$$

where $w_1 + w_2 + w_3 = 1$ are the positive weights,

$$\text{logit}(p_1) = \sum_{i=1}^{m} \beta_i z_i + \beta_{2m+1} z_{2m+1}$$

$$\text{logit}(p_2) = \sum_{i=m+1}^{2m} \beta_i z_i + \beta_{2m+1} z_{2m+2}$$

and

$$\text{logit}(p_3) = \eta + \beta_{2m+1} z_{2m+3}$$

for fixed covariates $\{z_i\}$. The linear predictor η is defined in the formula.

Note: β_{2m+1} is the *same* variable in the three expressions. The allowed range for m is $0 \le m \le 25$.

Link-function

The link-function is given as usual, and they are all equal.

Hyperparameters

The $2m + 1$ regression coefficients $\{\beta_i\}$ are treated as hyperparameters.

Specification

- `family="binomialmix"`
- Required arguments: A $n \times 2$ matrix Y with the observations and the number of trials s, $Y = (y, s)$, a $n \times (2m + 3)$ matrix Z with the covariates $Z = c(z_1, \ldots, z_{2m+3})$, and a $n \times 2$ matrix W with weights $W = (w_1, w_2)$. The `inla.mdata` is used as

```
inla.mdata(Y, Z, W) ~ ...
```

Hyperparameter spesification and default values

doc `Binomial mixture`

hyper

theta1

hyperid `56551`
name `beta1`
short.name `beta1`
output.name `beta1 for binomialmix observations`
output.name.intern `beta1 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta2

hyperid `56552`
name `beta2`
short.name `beta2`
output.name `beta2 for binomialmix observations`
output.name.intern `beta2 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta3

hyperid `56553`
name `beta3`
short.name `beta3`
output.name `beta3 for binomialmix observations`
output.name.intern `beta3 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta4

hyperid `56554`
name `beta4`
short.name `beta4`
output.name `beta4 for binomialmix observations`
output.name.intern `beta4 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`

param 0 100
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56555
name beta5
short.name beta5
output.name beta5 for binomialmix observations
output.name.intern beta5 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56556
name beta6
short.name beta6
output.name beta6 for binomialmix observations
output.name.intern beta6 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56557
name beta7
short.name beta7
output.name beta7 for binomialmix observations
output.name.intern beta7 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56558
name beta8
short.name beta8
output.name beta8 for binomialmix observations
output.name.intern beta8 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100

to.theta `function(x) x`
from.theta `function(x) x`

theta9

hyperid `56559`
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short.name `beta9`
output.name `beta9 for binomialmix observations`
output.name.intern `beta9 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta10

hyperid `56560`
name `beta10`
short.name `beta10`
output.name `beta10 for binomialmix observations`
output.name.intern `beta10 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta11

hyperid `56561`
name `beta11`
short.name `beta11`
output.name `beta11 for binomialmix observations`
output.name.intern `beta11 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta12

hyperid `56562`
name `beta12`
short.name `beta12`
output.name `beta12 for binomialmix observations`
output.name.intern `beta12 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`

from.theta `function(x) x`

theta13

hyperid `56563`
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short.name `beta13`
output.name `beta13 for binomialmix observations`
output.name.intern `beta13 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta14

hyperid `56564`
name `beta14`
short.name `beta14`
output.name `beta14 for binomialmix observations`
output.name.intern `beta14 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta15

hyperid `56565`
name `beta15`
short.name `beta15`
output.name `beta15 for binomialmix observations`
output.name.intern `beta15 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta16

hyperid `56566`
name `beta16`
short.name `beta16`
output.name `beta16 for binomialmix observations`
output.name.intern `beta16 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta17

hyperid 56567
name beta17
short.name beta17
output.name beta17 for binomialmix observations
output.name.intern beta17 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta18

hyperid 56568
name beta18
short.name beta18
output.name beta18 for binomialmix observations
output.name.intern beta18 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta19

hyperid 56569
name beta19
short.name beta19
output.name beta19 for binomialmix observations
output.name.intern beta19 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta20

hyperid 56570
name beta20
short.name beta20
output.name beta20 for binomialmix observations
output.name.intern beta20 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta21

hyperid 56571
name beta21
short.name beta21
output.name beta21 for binomialmix observations
output.name.intern beta21 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta22

hyperid 56572
name beta22
short.name beta22
output.name beta22 for binomialmix observations
output.name.intern beta22 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta23

hyperid 56573
name beta23
short.name beta23
output.name beta23 for binomialmix observations
output.name.intern beta23 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta24

hyperid 56574
name beta24
short.name beta24
output.name beta24 for binomialmix observations
output.name.intern beta24 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta25

hyperid 56575

name beta25
short.name beta25
output.name beta25 for binomialmix observations
output.name.intern beta25 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta26

hyperid 56576
name beta26
short.name beta26
output.name beta26 for binomialmix observations
output.name.intern beta26 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta27

hyperid 56577
name beta27
short.name beta27
output.name beta27 for binomialmix observations
output.name.intern beta27 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta28

hyperid 56578
name beta28
short.name beta28
output.name beta28 for binomialmix observations
output.name.intern beta28 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta29

hyperid 56579
name beta29

short.name beta29
output.name beta29 for binomialmix observations
output.name.intern beta29 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta30

hyperid 56580
name beta30
short.name beta30
output.name beta30 for binomialmix observations
output.name.intern beta30 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta31

hyperid 56581
name beta31
short.name beta31
output.name beta31 for binomialmix observations
output.name.intern beta31 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta32

hyperid 56582
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output.name beta32 for binomialmix observations
output.name.intern beta32 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta33

hyperid 56583
name beta33
short.name beta33

output.name beta33 for binomialmix observations
output.name.intern beta33 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta34

hyperid 56584
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short.name beta34
output.name beta34 for binomialmix observations
output.name.intern beta34 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta35

hyperid 56585
name beta35
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output.name beta35 for binomialmix observations
output.name.intern beta35 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta36

hyperid 56586
name beta36
short.name beta36
output.name beta36 for binomialmix observations
output.name.intern beta36 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta37

hyperid 56587
name beta37
short.name beta37
output.name beta37 for binomialmix observations

output.name.intern `beta37 for binomialmix observations`
initial `0`
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to.theta `function(x) x`
from.theta `function(x) x`

theta38

hyperid `56588`
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output.name.intern `beta38 for binomialmix observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta39

hyperid `56589`
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output.name.intern `beta39 for binomialmix observations`
initial `0`
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prior `normal`
param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta40

hyperid `56590`
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short.name `beta40`
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initial `0`
fixed `FALSE`
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param `0 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta41

hyperid `56591`
name `beta41`
short.name `beta41`
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output.name.intern `beta41 for binomialmix observations`

initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta42

hyperid 56592
name beta42
short.name beta42
output.name beta42 for binomialmix observations
output.name.intern beta42 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta43

hyperid 56593
name beta43
short.name beta43
output.name beta43 for binomialmix observations
output.name.intern beta43 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta44

hyperid 56594
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output.name beta44 for binomialmix observations
output.name.intern beta44 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta45

hyperid 56595
name beta45
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output.name beta45 for binomialmix observations
output.name.intern beta45 for binomialmix observations
initial 0

fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta46

hyperid 56596
name beta46
short.name beta46
output.name beta46 for binomialmix observations
output.name.intern beta46 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta47

hyperid 56597
name beta47
short.name beta47
output.name beta47 for binomialmix observations
output.name.intern beta47 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta48

hyperid 56598
name beta48
short.name beta48
output.name beta48 for binomialmix observations
output.name.intern beta48 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta49

hyperid 56599
name beta49
short.name beta49
output.name beta49 for binomialmix observations
output.name.intern beta49 for binomialmix observations
initial 0
fixed FALSE

prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta50

hyperid 56600
name beta50
short.name beta50
output.name beta50 for binomialmix observations
output.name.intern beta50 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta51

hyperid 56601
name beta51
short.name beta51
output.name beta51 for binomialmix observations
output.name.intern beta51 for binomialmix observations
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

survival FALSE

discrete TRUE

link default logit probit

pdf binomialmix

Example

Here is a simple example.

```
n <- 10^5
size <- sample(5:10, n, replace = TRUE)

m <- 5
m2 <- 2*m
beta.p1 <- rnorm(m, sd = 1/sqrt(m))
beta.p2 <- rnorm(m, sd = 1/sqrt(m))
beta.common <- rnorm(1, sd = 0.5)
beta <- c(beta.p1, beta.p2, beta.common)

Z <- matrix(NA, n, m2+3)
W <- matrix(NA, n, 2)
Y <- matrix(NA, n, 2)

x <- rnorm(n, sd = 0.5)
xx <- rnorm(n, sd = 0.5)
eta <- numeric(n)

for (i in 1:n) {
    Z[i, ] <- rnorm(m2+3)
    w <- c(rbeta(2, 1, 10), rbeta(1, 10, 1))
    w <- w/sum(w)
    W[i, ] <- w[1:2]

    p1 <- inla.link.invlogit(sum(beta.p1 * Z[i, seq_len(m)]) + beta.common * Z[i, m2+1])
    p2 <- inla.link.invlogit(sum(beta.p2 * Z[i, m + seq_len(m)]) + beta.common * Z[i, m2+2])

    eta[i] <-  1 + x[i] + xx[i] + beta.common * Z[i, m2+3]
    p3 <- inla.link.invlogit(eta[i])

    p <- w[1] * p1 + w[2] * p2 + w[3] * p3
    Y[i, ] <- c(rbinom(1, size = size[i], prob = p), size[i])
}

r <- inla(inla.mdata(Y, Z, W) ~ 1 + x + xx,
          family = "binomialmix",
          data = list(Y = Y, Z = Z, W = W, x = x, xx = xx),
          verbose = TRUE,
          control.inla = list(int.strategy = "eb"))
## r <- inla.rerun(r)

print(round(dig = 4, cbind(estimate = r$summary.fixed[,"mean"], true = 1)))
print(round(dig = 4, cbind(estimate =  r$summary.hyperpar[,"mean"], true = beta)))
```